

MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Internship

PERSONAL CYBERSECURITY LAB

Task – 2

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Maincrafts Technology**

Task Objective

To build a secure and isolated cybersecurity laboratory environment using Kali Linux, Docker, and OWASP Juice Shop for practicing web application security testing, network analysis, and cybersecurity fundamentals.

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Abstract

This report documents the creation of a personal cybersecurity laboratory using Oracle VirtualBox, Kali Linux, Docker, and OWASP Juice Shop. The purpose of the lab was to establish a safe and isolated environment for cybersecurity learning and experimentation. Kali Linux was configured as the primary testing machine, while OWASP Juice Shop was deployed as a vulnerable target application using Docker containers. Connectivity validation was performed using IP configuration analysis, ping testing, and Nmap scanning. Web traffic interception was carried out using Burp Suite, and network packets were analyzed using Wireshark. The successful completion of these activities demonstrates the effectiveness of virtualization and containerization technologies in building practical cybersecurity training environments.

Introduction

Cybersecurity professionals frequently require isolated environments for practicing penetration testing, vulnerability assessment, network monitoring, and web application security testing. Creating such environments on production systems can be risky; therefore, virtualization technologies are widely used to provide secure and controlled testing platforms.

This project focuses on building a personal cybersecurity laboratory using Oracle VirtualBox and Kali Linux. OWASP Juice Shop was deployed as a deliberately vulnerable web application to serve as a testing target. Various security tools including Docker, Nmap, Burp Suite, and Wireshark were utilized to validate network connectivity and monitor traffic within the laboratory environment.

The completed lab serves as a foundation for future cybersecurity activities such as vulnerability assessment, web application testing, and network traffic analysis.

Objectives

The objectives of this task were:

- To create an isolated cybersecurity laboratory environment.
- To configure VirtualBox networking using NAT and Host-Only networking.
- To deploy Kali Linux as the attacker machine.
- To deploy OWASP Juice Shop as a vulnerable web application.
- To validate network connectivity using Ping and Nmap.
- To monitor HTTP traffic using Burp Suite.
- To analyze network packets using Wireshark.
- To establish a reusable platform for cybersecurity practice.

System Configuration

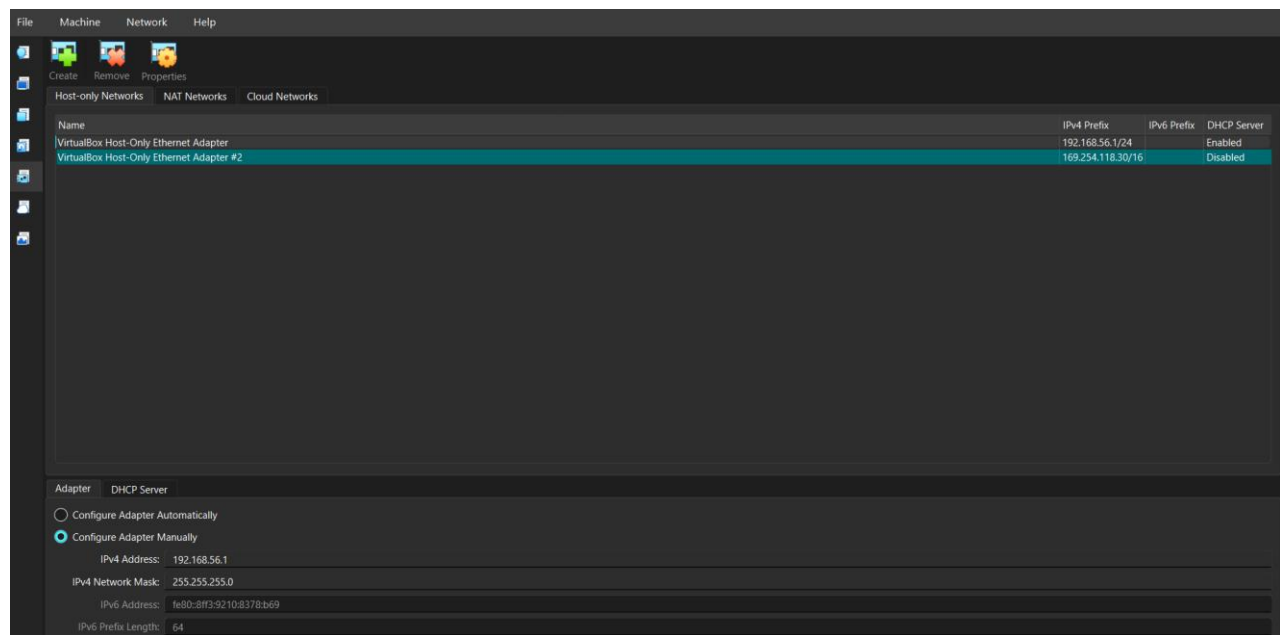
Component	Configuration
Virtualization Software	Oracle VirtualBox
Operating System	Kali Linux
RAM Assigned	4096 MB
CPUs Assigned	4
Network Type	NAT + Host-Only
Container Platform	Docker
Vulnerable Application	OWASP Juice Shop

Methodology

Step 1: Host-Only Network Configuration

A Host-Only Network Adapter was created in VirtualBox to establish isolated communication within the cybersecurity laboratory.

Figure 1: Host-Only Network Configuration



Step 2: Kali Linux Virtual Machine Setup

The Kali Linux virtual machine was configured with 4096 MB RAM and 4 virtual CPUs to ensure stable operation.

Figure 2: Kali Linux Memory Configuration

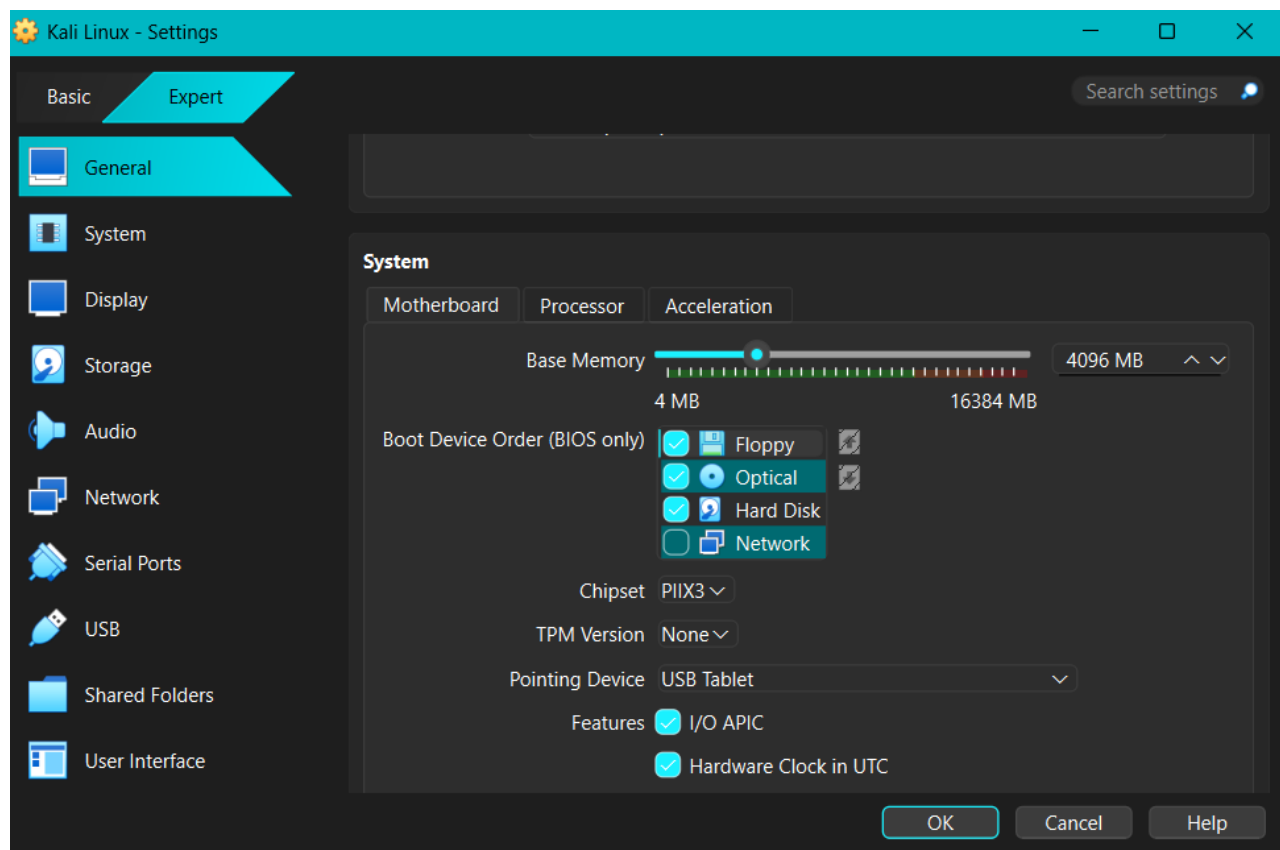
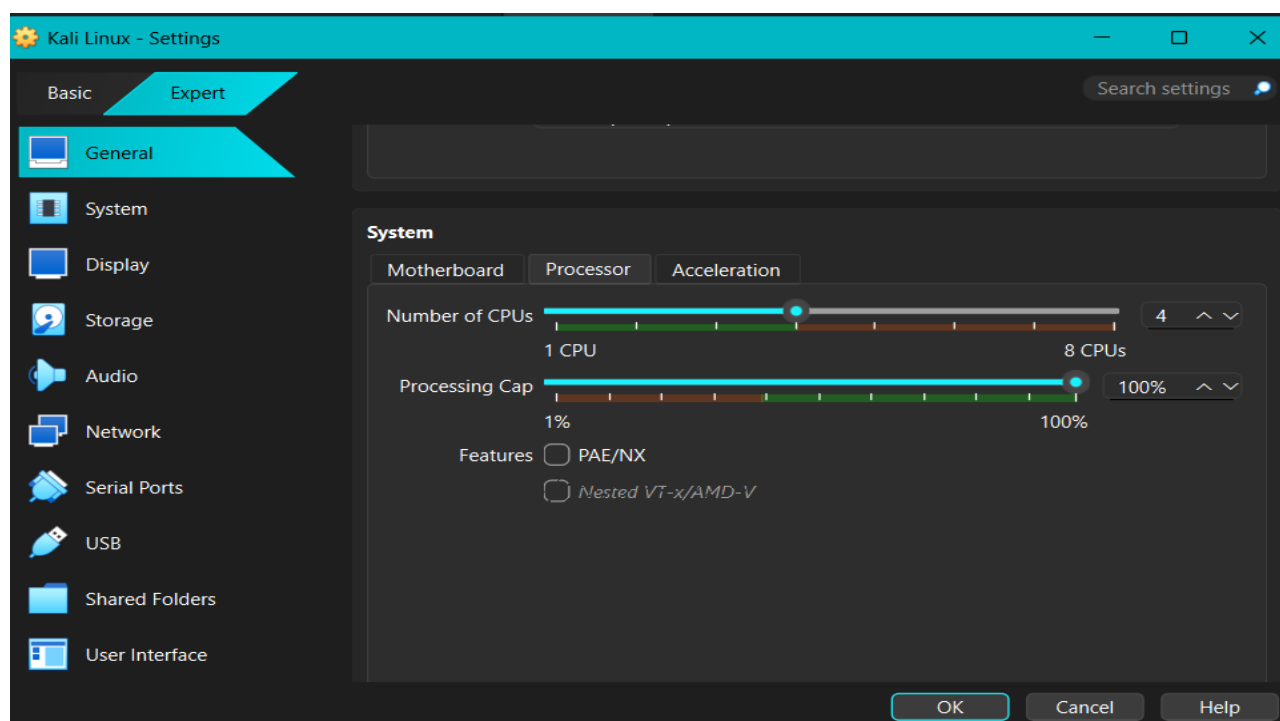


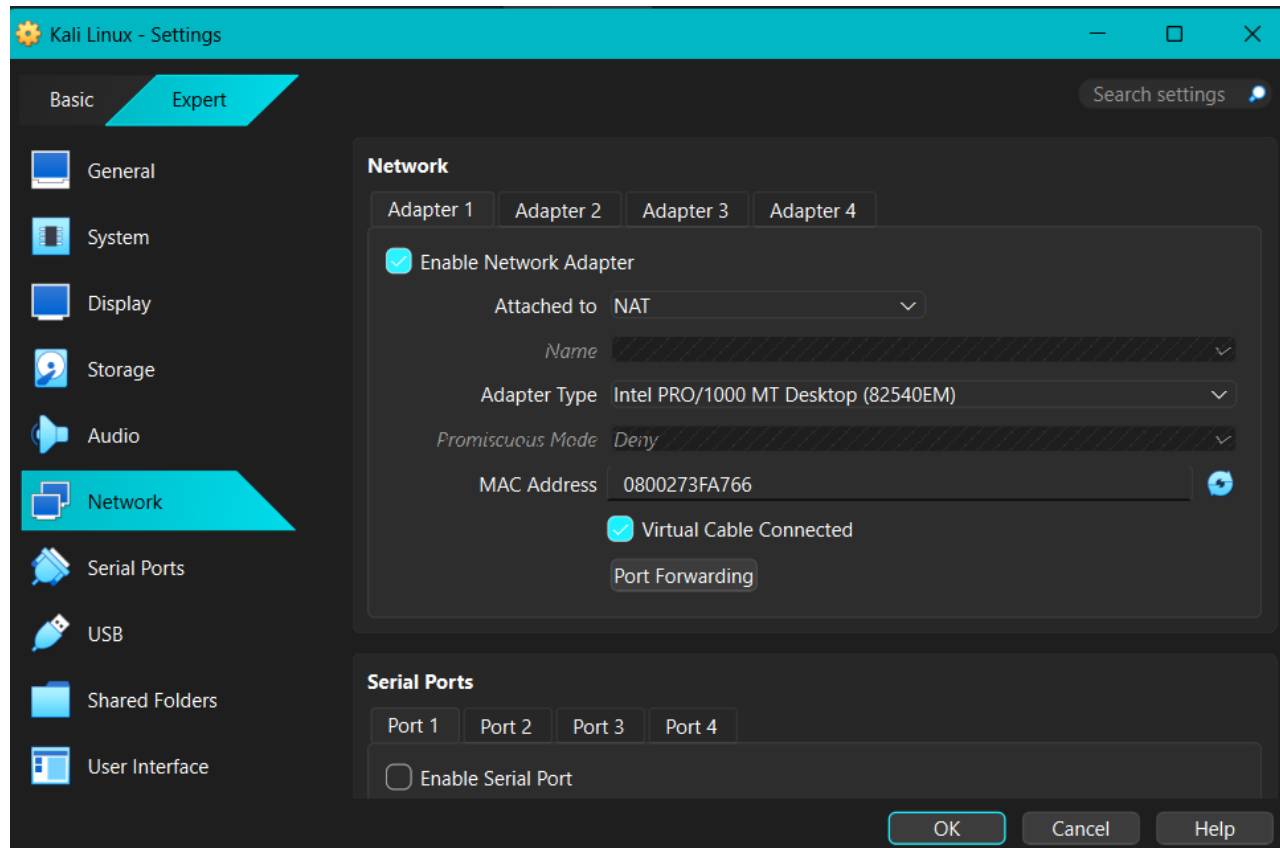
Figure 3: Kali Linux Processor Configuration



Step 3: Network Adapter Configuration

The virtual machine network adapter was configured using NAT networking to provide internet connectivity.

Figure 4: Kali Linux Network Configuration



Step 4: Laboratory Workspace Creation

Dedicated folders were created to store scans, notes, packet captures, and screenshots.

Figure 5: Laboratory Directory Structure



Step 5: Docker Installation and Verification

Docker was installed and verified successfully within Kali Linux.

Figure 6: Docker Installation Verification

```
(peter@kali) ~/Documents
$ sudo systemctl enable --now docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker

(peter@kali) ~/Documents
$ docker --version
Docker version 20.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed

(peter@kali) ~/Documents
```

Step 6: OWASP Juice Shop Deployment

OWASP Juice Shop was downloaded from Docker Hub and deployed using a Docker container.

Figure 7: OWASP Juice Shop Deployment

```
(peter@kali) ~/Documents
$ docker pull bkimminich/juice-shop
Using default tag: latest
latest: Pulling from bkimminich/juice-shop
470c5d08b12: Pull complete
c172f21841df: Pull complete
99515e7b4d35: Pull complete
990a92a91a2: Pull complete
d0a3b09accac: Pull complete
2780920e5d0f: Pull complete
7c12809b777b: Pull complete
321aac7345a8: Pull complete
52630fc75a18: Pull complete
d064b2d0177: Pull complete
b839dfaa016: Pull complete
ebddc55facdc: Pull complete
bdf077f63bf6: Pull complete
24d7ad16272: Pull complete
40008157d8d2: Pull complete
b0892e29291: Pull complete
cac2a09105cb: Pull complete
d0049bdf3bee: Pull complete
7cc2bdf680cf: Pull complete
8224d91da7bb: Pull complete
1a4be5562d92: Pull complete
309faeb031be: Pull complete
5e22571f04b1: Pull complete
8880d8eb208: Pull complete
Digest: sha256:5d572829141b32e065edefc0b09ca88567206ff7134ebc3181ab7ac9e9ed4056
Status: Downloaded newer image for bkimminich/juice-shop:latest
docker.io/bkimminich/juice-shop:latest

(peter@kali) ~/Documents
$ sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
5eb97f0a99fca20d47b29644dcf2710d6f1d85ea341f9bc50b57f4c49035f

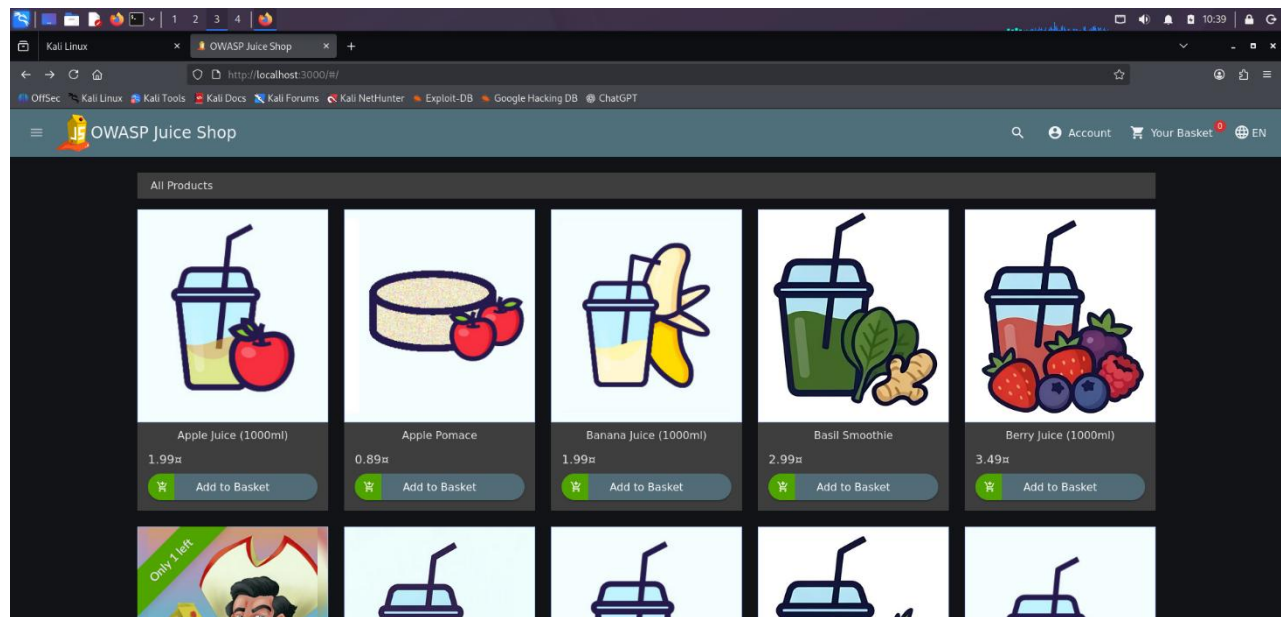
(peter@kali) ~/Documents
$ sudo docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                               NAMES
5eb97f0a99fc   bkimminich/juice-shop  "/nodejs/bin/node /j-..."  13 seconds ago Up 12 seconds  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   juice

(peter@kali) ~/Documents
```

Step 7: Application Verification

The deployed application was successfully accessed through a web browser.

Figure 8: OWASP Juice Shop Homepage



Step 8: Network Validation

IP configuration details were collected and connectivity testing was performed using Ping and Nmap.

Figure 9: IP Configuration

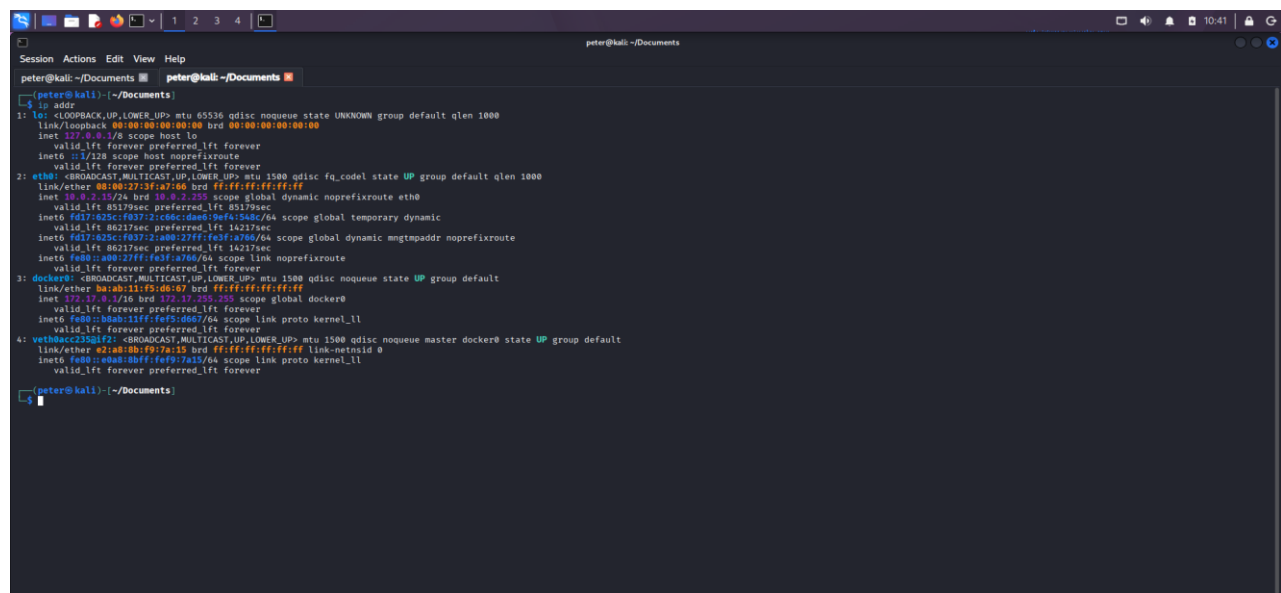


Figure 10: Ping Validation



Figure 11: Nmap Discovery

```
(peter@kali) (~/.Documents)
$ sudo nmap -sn 172.17.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-19 10:44 +0100
Nmap scan report for 172.17.0.1
Host is up.
Nmap done: 1 IP address (1 host up) scanned in 2.60 seconds

(peter@kali) (~/.Documents)
$ sudo nmap -sn -v 172.17.0.1
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.4 openssl-3.6.2 libssh2-1.11.1 lib-1.3.1 libpcap-10.46 libcap-1.10.0 nmap-libndm-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(peter@kali) (~/.Documents)
```

Step 9: Traffic Monitoring and Analysis

Burp Suite was used to intercept HTTP requests while Wireshark was used to capture and analyze network packets.

Figure 12: Burp Suite Interception

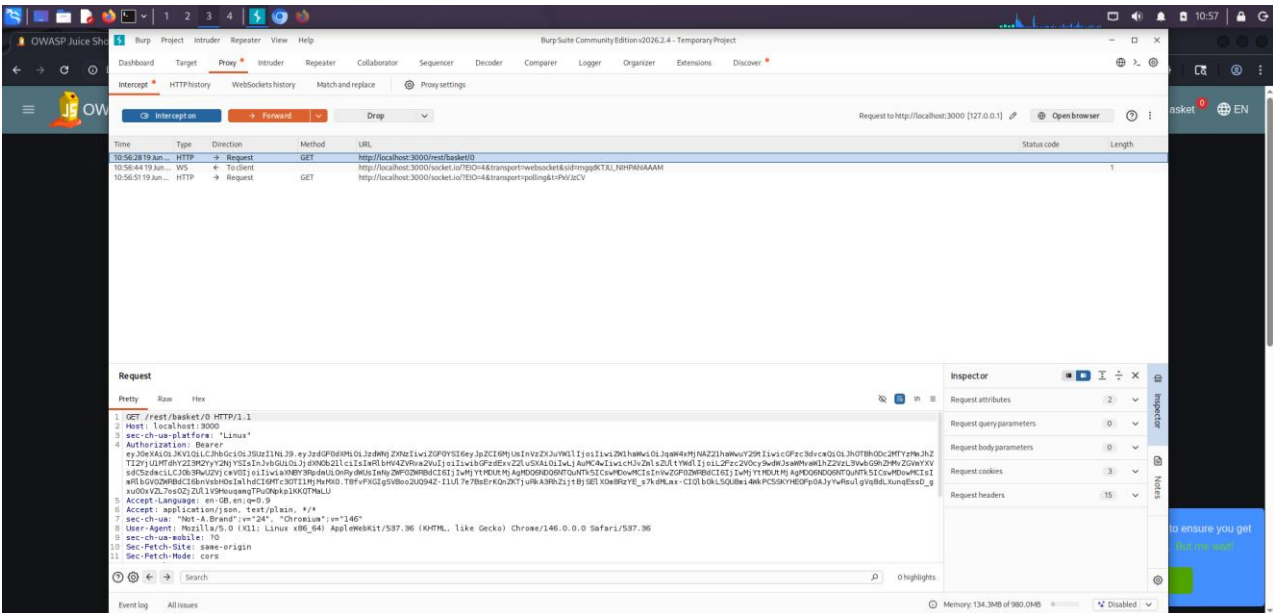
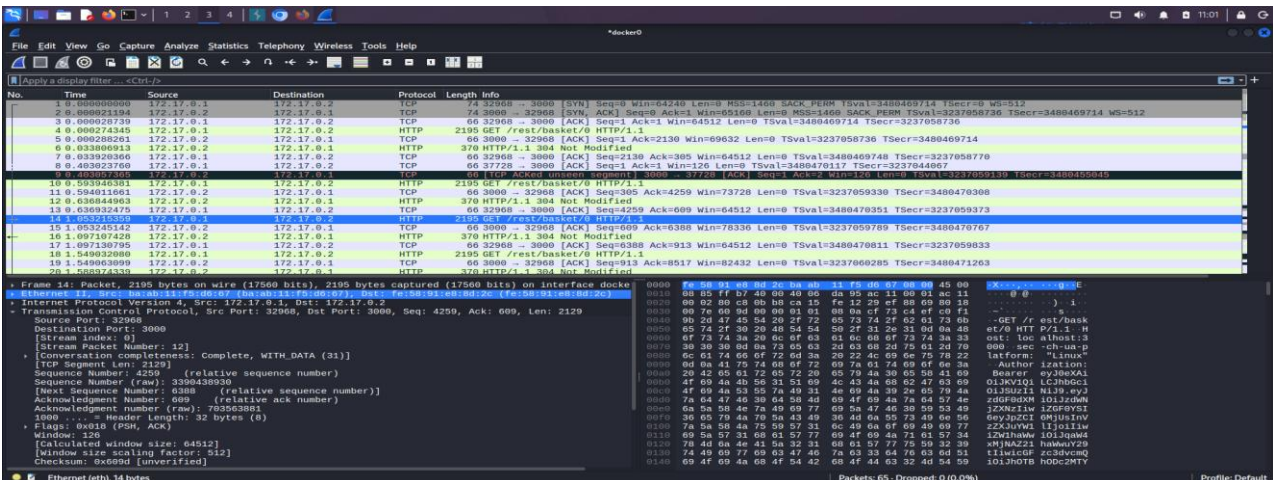


Figure 13: Wireshark Packet Capture



Results and Analysis

Table 1: Virtual Machine Configuration

Parameter	Value
RAM	4096 MB
CPU	4 vCPUs
Platform	VirtualBox
Network	NAT + Host-Only

Table 2: Connectivity Validation Results

Test	Result
IP Configuration	Successful
Ping Test	Successful
Packet Loss	0%
Nmap Discovery	Successful

Table 3: Security Tool Validation

Tool	Purpose	Status
Docker	Container Deployment	Successful
OWASP Juice Shop	Vulnerable Web Application	Running
Nmap	Host Discovery	Successful
Burp Suite	HTTP Traffic Interception	Successful
Wireshark	Packet Analysis	Successful

Discussion

The cybersecurity laboratory was successfully implemented using Oracle VirtualBox and Kali Linux. Docker enabled efficient deployment of OWASP Juice Shop as a vulnerable target application. Network validation confirmed successful communication between system components, while Nmap verified host availability.

Burp Suite demonstrated the ability to intercept HTTP requests generated by the application, and Wireshark successfully captured network traffic for analysis. The experiment highlights the effectiveness of virtualization and containerization technologies in creating practical environments for cybersecurity learning and testing.

Conclusion

The objective of building a personal cybersecurity laboratory was successfully achieved. Kali Linux was configured as the primary testing environment, and OWASP Juice Shop was deployed as a vulnerable web application using Docker. Network validation, traffic interception, and packet analysis were successfully performed using industry-standard cybersecurity tools.

The completed environment provides a secure platform for future cybersecurity exercises, including vulnerability assessment, web application testing, and network monitoring.

References

- [1] Oracle VirtualBox Documentation.
- [2] Kali Linux Official Documentation.
- [3] Docker Official Documentation.
- [4] OWASP Juice Shop Documentation.
- [5] Nmap Reference Guide.
- [6] Burp Suite Documentation.
- [7] Wireshark User Guide.